

Project #11: Groove McChord Field Main Runway

Project Title: Runway Pavement Grooving at McChord Airfield

1. Project Overview

The Contractor shall provide all labor, equipment, materials, and supervision necessary to perform pavement grooving on a specific portion of the active runway at McChord Airfield. The objective of this project is to enhance aircraft braking performance and reduce hydroplaning risk by cutting grooves into the existing concrete pavement surface.

2. Location of Work

The work shall be performed at the north end of the primary concrete runway. The specific area to be grooved is the first 172 linear feet of the ungrooved concrete section, as detailed in the attached "Runway Grooving Diagram.pdf". The Contractor is responsible for verifying all field measurements prior to commencing work.

3. Groove Specifications

The Contractor shall cut grooves according to one of the two profiles specified below. The final groove profile selection (Standard Square or Trapezoidal) will be confirmed by the Contracting Officer prior to the start of work.

Option A: Standard Square Groove Profile

Specification	Dimension	Tolerance
Width	1/4 inch (6 mm)	+1/16 inch, -0 inch
Depth	1/4 inch (6 mm)	± 1/16 inch
Center-to-Center Spacing	1 1/2 inches (38 mm)	-1/8 inch, +0 inch

Option B: Trapezoidal Groove Profile

Specification	Dimension	Tolerance
Width at Top	1/2 inch (12 mm)	+1/16 inch, -0 inch
Width at Bottom	1/4 inch (6 mm)	+1/16 inch, -0 inch
Depth	1/4 inch (6 mm)	± 1/16 inch

Center-to-Center Spacing	2 1/4 inches (57 mm)	-1/8 inch, +0 inch
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4. Execution and Construction Requirements

- **Pavement Preparation:** Prior to starting, the Contractor shall inspect the designated grooving area for any existing pavement damage, including spalls, cracks, or significant surface irregularities. Any such defects must be reported to the Agreements Officer. Grooving shall not commence until these defects have been addressed by the appropriate authority.
- **Groove Orientation:** All grooves shall be cut uniformly and continuously, transverse (perpendicular) to the runway centerline.
- **Alignment Tolerance:** The horizontal alignment of the grooves shall not vary by more than 1-1/2 inches in any 75-foot length.
- **Cleanup and Environmental:** The Contractor shall implement a continuous cleanup operation. All cutting slurry, water, and debris must be collected and removed from the pavement surface immediately. The waste material shall be legally disposed of off-site and must not be permitted to enter airfield drainage systems.
- **Foreign Object Debris (FOD):** The work area must be kept free of FOD at all times. The runway surface must be thoroughly cleaned and inspected for FOD prior to returning it to operational status.

5. Quality Control and Acceptance

- At least 90% of all grooves, for their full length, must have a minimum depth of 3/16 inch.
- At least 60% of all grooves, for their full length, must meet the full specified depth of 1/4 inch.
- Final acceptance of the work is subject to inspection and verification of all specifications by the Government's designated representative.

6. Work Constraints & Coordination

- **Airfield Operations:** This is an active airfield. All work must be closely coordinated with Airfield Operations and the Contracting Officer. The Contractor is responsible for adhering to all security and access requirements.

- Work Hours: Work hours will be strictly limited to minimize impact on flight operations and will be determined by the Contracting Officer. The Contractor should anticipate that work may be restricted to nighttime hours.
- Weather Limitations: Grooving operations are not permitted when ambient temperatures are at or below freezing (32°F / 0°C), or if freezing conditions are expected that would prevent the proper removal of slurry and debris.

7. Reference Documents

The Contractor is expected to be familiar with and perform all work in accordance with the following documents:

- Runway Grooving Diagram.pdf (Provided)
- Unified Facilities Guide Specification (UFGS) 32 01 18.71, Grooving of Airfield Paving

